Linear Models for Classification in Machine Learning

classify the data with linear boundary.

1. Introduction to Classification

Classification is a supervised learning task where the goal is to predict discrete class labels for input data. Unlike regression, which predicts continuous values, classification assigns inputs to predefined categories or classes.

Key Concepts:

- **Binary Classification**: Two classes (e.g., spam/not spam, disease/no disease)
- **Multi-class Classification**: More than two classes (e.g., digit recognition 0-9)
- **Features**: Input variables $\mathbf{x} = (x_1, x_2, \dots, x_D)^T$
- **Target**: Class label $y \in \{C_1, C_2, \dots, C_K\}$

Linear models for classification construct decision boundaries that are linear functions of the input features. Despite their simplicity, these models are powerful, interpretable, and form the foundation for understanding more complex classification methods[1].

2. Decision Boundaries and Linear Discriminants

2.1 Linear Decision Boundary

A linear decision boundary in D-dimensional input space is defined by:

$$y(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + w_0 = \sum_{i=1}^{D} w_i x_i + w_0 = 0$$

where:

- $\mathbf{w} = (w_1, w_2, \dots, w_D)^T$ is the weight vector (normal to the decision surface)
- w_0 is the bias term (also called threshold or intercept)
- $\mathbf{x} = (x_1, x_2, \dots, x_D)^T$ is the input feature vector
- D is the dimensionality of the input space

Detailed Explanation:

The function $y(\mathbf{x})$ is a linear combination of the input features. The decision boundary is the set of points where $y(\mathbf{x}) = 0$, which forms a hyperplane in D-dimensional space:

- In 2D: the boundary is a line

- In 3D: the boundary is a plane
- In higher dimensions: the boundary is a hyperplane

Classification Rule:

$$\text{Class} = \begin{cases} C_1 & \text{if } y(\mathbf{x}) \geq 0 \\ C_2 & \text{if } y(\mathbf{x}) < 0 \end{cases}$$

The sign of $y(\mathbf{x})$ determines which side of the hyperplane the point lies on. Points exactly on the boundary have $y(\mathbf{x}) = 0$.

Geometric Properties:

1. **Normal Vector**: The weight vector $\mathbf{w}$ is perpendicular (normal) to the decision boundary.

2. **Bias Term Effect**: The bias w_0 shifts the boundary parallel to itself without changing its orientation. Specifically:

    - If $w_0 = 0$, the boundary passes through the origin (centre)
    - If $w_0 > 0$, the boundary shifts in the direction of $-\mathbf{w}$
    - If $w_0 < 0$, the boundary shifts in the direction of $+\mathbf{w}$

3. **Homogeneous Coordinates**: We can incorporate the bias by augmenting the weight vector and input: $\tilde{\mathbf{w}} = (w_0, w_1, \ldots, w_D)^T$ and $\tilde{\mathbf{x}} = (1, x_1, \ldots, x_D)^T$, giving $y(\mathbf{x}) = \tilde{\mathbf{w}}^T\tilde{\mathbf{x}}$.

2.2 Geometric Interpretation

The perpendicular distance from a point $\mathbf{x}$ to the decision boundary is:

$$r = \frac{y(\mathbf{x})}{\|\mathbf{w}\|} = \frac{|\mathbf{w}^T\mathbf{x} + w_0|}{\sqrt{w_1^2 + w_2^2 + \cdots + w_D^2}}$$

Derivation:

Consider a point $\mathbf{x}$ and its projection $\mathbf{x}_\perp$ onto the decision boundary. The vector from $\mathbf{x}_\perp$ to $\mathbf{x}$ is parallel to $\mathbf{w}$:

$$\mathbf{x} = \mathbf{x}_\perp + r\frac{\mathbf{w}}{\|\mathbf{w}\|}$$

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Since $\mathbf{x}_\perp$ is on the boundary, $\mathbf{w}^T\mathbf{x}_\perp + w_0 = 0$. Substituting:

$$y(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + w_0 = \mathbf{w}^T\left(\mathbf{x}_\perp + r\frac{\mathbf{w}}{\|\mathbf{w}\|}\right) + w_0$$

$$= \mathbf{w}^T\mathbf{x}_\perp + w_0 + r\frac{\mathbf{w}^T\mathbf{w}}{\|\mathbf{w}\|} = 0 + r\frac{\|\mathbf{w}\|^2}{\|\mathbf{w}\|} = r\|\mathbf{w}\|$$

Therefore:

$$r = \frac{y(\mathbf{x})}{||\mathbf{w}||}$$

Interpretation:

- If $y(\mathbf{x}) > 0$: point is on the positive side (direction of $\mathbf{w}$), distance is $r = \frac{y(\mathbf{x})}{||\mathbf{w}||}$
- If $y(\mathbf{x}) < 0$: point is on the negative side, distance is $r = \frac{-y(\mathbf{x})}{||\mathbf{w}||}$
- If $y(\mathbf{x}) = 0$: point is on the boundary, distance is 0

The signed distance (with sign indicating which side) is $\frac{y(\mathbf{x})}{||\mathbf{w}||}$.

Example Problem 2.1: Find the distance from point $\mathbf{x} = (3,4)^T$ to the decision boundary defined by $2x_1 + x_2 - 5 = 0$.

Solution:

Given: $\mathbf{x} = (3,4)^T$, decision boundary: $2x_1 + x_2 - 5 = 0$

Weight vector: $\mathbf{w} = (2,1)^T$, bias: $w_0 = -5$

Step 1: Compute $y(\mathbf{x})$:

$$y(\mathbf{x}) = 2(3) + 1(4) - 5 = 6 + 4 - 5 = 5$$

Step 2: Compute $||\mathbf{w}||$:

$$||\mathbf{w}|| = \sqrt{2^2 + 1^2} = \sqrt{4 + 1} = \sqrt{5} \approx 2.236$$

Step 3: Compute the signed distance:

$$r = \frac{y(\mathbf{x})}{||\mathbf{w}||} = \frac{5}{\sqrt{5}} = \frac{5\sqrt{5}}{5} = \sqrt{5} \approx 2.236$$

Conclusion: The point $\mathbf{x} = (3,4)^T$ is at distance $\sqrt{5} \approx 2.236$ units from the boundary, on the positive side (Class C_1). Since $y(\mathbf{x}) = 5 > 0$, the point would be classified as C_1.

Example Problem 2.2: Consider the decision boundary $x_1 - 2x_2 + 3 = 0$. Find the perpendicular distance from the following points and classify them:
(a) $\mathbf{x}_a = (1,1)^T$
(b) $\mathbf{x}_b = (-1,2)^T$
(c) $\mathbf{x}_c = (3,3)^T$

Solution:

Given: $\mathbf{w} = (1,-2)^T$, $w_0 = 3$, $||\mathbf{w}|| = \sqrt{1^2 + (-2)^2} = \sqrt{5}$

(a) For $\mathbf{x}_a = (1,1)^T$:

$$y(\mathbf{x}_a) = 1(1) - 2(1) + 3 = 1 - 2 + 3 = 2$$

$$r_a = \frac{2}{\sqrt{5}} = \frac{2\sqrt{5}}{5} \approx 0.894$$

Classification: C_1 (since $y(\mathbf{x}_a) > 0$), distance ≈ 0.894 units.

(b) For $\mathbf{x}_b = (-1,2)^T$:

$$y(\mathbf{x}_b) = 1(-1) - 2(2) + 3 = -1 - 4 + 3 = -2$$

$$r_b = \frac{-2}{\sqrt{5}} = \frac{-2\sqrt{5}}{5} \approx -0.894$$

Classification: C_2 (since $y(\mathbf{x}_b) < 0$), distance ≈ 0.894 units (on negative side).

(c) For $\mathbf{x}_c = (3,3)^T$:

$$y(\mathbf{x}_c) = 1(3) - 2(3) + 3 = 3 - 6 + 3 = 0$$

$$r_c = \frac{0}{\sqrt{5}} = 0$$

Classification: On the boundary (exactly on the decision surface), distance = 0.

Example Problem 2.3: Two parallel decision boundaries are defined by $2x_1 + x_2 - 4 = 0$ and $2x_1 + x_2 + 2 = 0$. Find the perpendicular distance between these two boundaries.

Solution:

Both boundaries have the same weight vector $\mathbf{w} = (2,1)^T$, so they are parallel.

Method 1 (Using any point on one boundary):

Take a point on the first boundary, say $\mathbf{x}_0 = (2,0)^T$ (verify: $2(2) + 0 - 4 = 0$ ✓).

Find its distance to the second boundary $2x_1 + x_2 + 2 = 0$:

$$y(\mathbf{x}_0) = 2(2) + 1(0) + 2 = 4 + 0 + 2 = 6$$

$$||\mathbf{w}|| = \sqrt{2^2 + 1^2} = \sqrt{5}$$

$$d = \frac{|y(\mathbf{x}_0)|}{||\mathbf{w}||} = \frac{6}{\sqrt{5}} = \frac{6\sqrt{5}}{5} \approx 2.683$$

Method 2 (Using bias difference):

For parallel hyperplanes with the same **w** but different biases $w_{0,1}$ and $w_{0,2}$:

$$d = \frac{|w_{0,1} - w_{0,2}|}{||\mathbf{w}||}$$

Here: $w_{0,1} = -4$, $w_{0,2} = 2$

$$d = \frac{|-4-2|}{\sqrt{5}} = \frac{6}{\sqrt{5}} = \frac{6\sqrt{5}}{5} \approx 2.683$$

Conclusion: The perpendicular distance between the two parallel boundaries is $\frac{6\sqrt{5}}{5} \approx 2.683$ units.

Example Problem 2.4: A decision boundary is defined by $3x_1 - 4x_2 + 10 = 0$. Find:
(a) A unit normal vector to the boundary
(b) The distance from the origin to the boundary
(c) A point on the boundary closest to the origin

Solution:

Given: $\mathbf{w} = (3, -4)^T$, $w_0 = 10$

(a) Unit normal vector:

$$\|\mathbf{w}\| = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = 5$$

$$\hat{\mathbf{w}} = \frac{\mathbf{w}}{\|\mathbf{w}\|} = \frac{1}{5}(3, -4)^T = (0.6, -0.8)^T$$

(b) Distance from origin $\mathbf{x}_0 = (0,0)^T$:

$$y(\mathbf{x}_0) = 3(0) - 4(0) + 10 = 10$$

$$r = \frac{|y(\mathbf{x}_0)|}{\|\mathbf{w}\|} = \frac{10}{5} = 2$$

Distance from origin to boundary: 2 units.

(c) Closest point on boundary:

The closest point is along the normal direction from the origin:

$$\mathbf{x}_{\text{closest}} = \mathbf{x}_0 - r\hat{\mathbf{w}} = (0,0)^T - 2(0.6, -0.8)^T = (-1.2, 1.6)^T$$

Verification: Check if this point is on the boundary:

$$3(-1.2) - 4(1.6) + 10 = -3.6 - 6.4 + 10 = 0 \checkmark$$

The closest point on the boundary to the origin is $(-1.2, 1.6)^T$.

Example Problem 2.5: Given three points $\mathbf{x}_1 = (1,2)^T$, $\mathbf{x}_2 = (2,3)^T$, $\mathbf{x}_3 = (4,1)^T$, and a decision boundary $x_1 + x_2 - 4 = 0$. Which point is farthest from the boundary?

Solution:

Given: $\mathbf{w} = (1,1)^T$, $w_0 = -4$, $\|\mathbf{w}\| = \sqrt{1^2 + 1^2} = \sqrt{2}$

For $\mathbf{x}_1 = (1,2)^T$:

$$y(\mathbf{x}_1) = 1 + 2 - 4 = -1$$

$$r_1 = \frac{|-1|}{\sqrt{2}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \approx 0.707$$

For $\mathbf{x}_2 = (2,3)^T$:

$$y(\mathbf{x}_2) = 2 + 3 - 4 = 1$$

$$r_2 = \frac{|1|}{\sqrt{2}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \approx 0.707$$

For $\mathbf{x}_3 = (4,1)^T$:

$$y(\mathbf{x}_3) = 4 + 1 - 4 = 1$$

$$r_3 = \frac{|1|}{\sqrt{2}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \approx 0.707$$

Conclusion: Interestingly, all three points are equidistant from the boundary at $\frac{\sqrt{2}}{2} \approx 0.707$ units. However, $\mathbf{x}_1$ is on the negative side (Class C_2), while $\mathbf{x}_2$ and $\mathbf{x}_3$ are on the positive side (Class C_1).